

CRISPR-Cas9 Endonuclease Mechanisms and Genome Editing Precision

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1. Bacterial Adaptive Immunity and Guide RNA Hybridization

The Type II CRISPR-Cas9 system from *Streptococcus pyogenes* uses a dual RNA guide consisting of CRISPR RNA (crRNA) and trans-activating crRNA (tracrRNA), engineered into a single guide RNA (sgRNA).

2. Protospacer Adjacent Motif (PAM) Recognition and Double-Strand Breaks

Target DNA cleavage requires a canonical 5'-NGG-3' PAM sequence immediately adjacent to the 20-nucleotide protospacer.

Cas9 catalyzes double-stranded breaks via two endonuclease domains:

- RuvC domain: cleaves the non-target DNA strand
- HNH domain: cleaves the target DNA strand complementary to the gRNA

3. DNA Repair Pathways: NHEJ vs HDR

Table 1: Comparison of Cas9 repair outcomes.

Pathway	Template Required	Error Rate	Primary Outcome
NHEJ	None	High (InDels)	Gene Knockout / Frameshift
HDR	Exogenous Donor	High Fidelity	Precision Gene Knock-in / Point Mutation

Figure 1 (Page 1): Crystal structure depiction of Cas9-sgRNA ribonucleoprotein complex bound to DNA target.